

Introduction to Computer Aided Design (3D Modeling)

Prototyping Skills (BME290L)

Mark L. Palmeri, M.D., Ph.D.

January 17, 2023

Outline

General Workflow

Modifying Edges

Common Types

Why?

Holes

Screws

Mechanical Drawings

Mechanical Design Considerations

Demo

Lab

OnShape



- ▶ We will be using OnShape, a cloud-based CAD package with a similar workflow to SolidWorks.
- ▶ You have been automatically registered to join Duke's OnShape team (<https://duke.onshape.com>).

Tutorials

- ▶ OnShape
 - ▶ Introduction to Sketching
 - ▶ Introduction to Part Design
 - ▶ Introduction to 2D Drawings
- ▶ Enclosure Design for 3D Printing
- ▶ How to assemble 3D printed parts with threaded fasteners
- ▶ How to design snap-fit joints for 3D printing

General Workflow

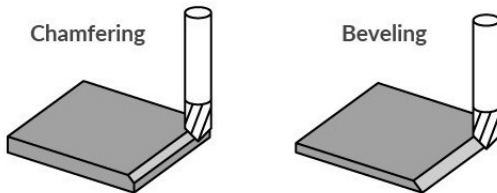
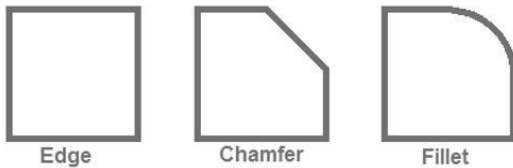
1. Sketch 2D profiles (dimensioned).
2. Use constraints to align sketch features to one another.
3. Extend 2D profiles to 3D parts.
 - ▶ Extrude (new, add, subtract, intersection)
 - ▶ Revolve
4. Utilize part symmetry and patterns to reduce manual effort.
5. Define new sketch planes on part references that “make sense”.
6. Assemble multiple parts.

Practice Makes Perfect

- ▶ There are many ways to create the same parts. Some are more amenable to future modification than others; those aren't always the fastest to create from scratch.
- ▶ Experience is highly valuable!

Modifying Edges

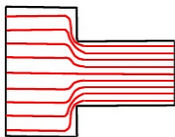
- ▶ Fillet: rounded corner / edge
- ▶ Chamfer: sloped / angled corner / edge



Why?

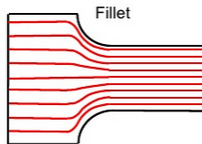
Why modify edges?

- ▶ Reduce stress concentrations (fillet > chamfer).
- ▶ Reduce sharp edges.
- ▶ Ease assembly (potential at cost to manufacturing).



Abrupt change

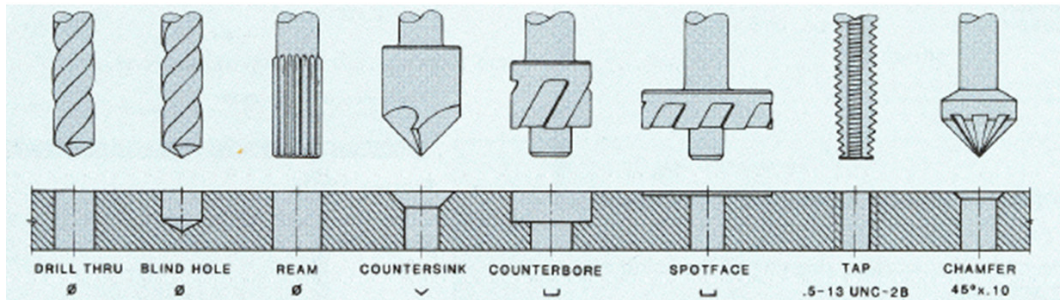
Stress "flow lines" crowd together causing high stress concentration in transition zone



Smooth change

"Flow lines" more evenly distributed causing lower stress concentration in transition zone

Types of Holes



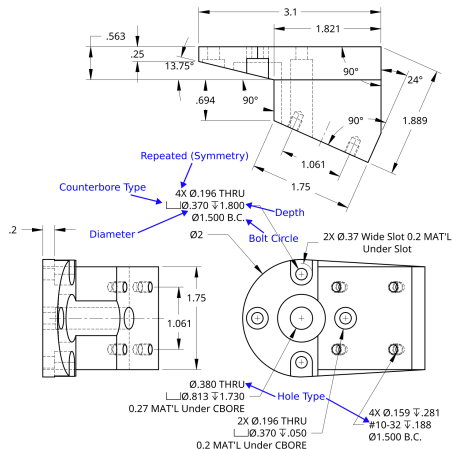
CAD software has a specific Hole tool that can be very handy. Specify diameters and threading based on ANSI and ISO standards (e.g., ANSI #10-32 bolt or ISO M5-0.8). Additionally, a table of holes can be generated on mechanical drawings.

Types of Screws



Mechanical Drawings

- ▶ Multiview / orthographic projection
- ▶ ISO 8015 (“Geometrical product specifications (GPS) — Fundamentals — Concepts, principles and rules”)
- ▶ Dimension callouts / values should not overlap any part lines!



Mechanical Design Considerations

- ▶ Purpose
 - ▶ Protection (water, debris?)
 - ▶ Holding components in relation to each other
 - ▶ Mounting
- ▶ Loading
 - ▶ Shipping
 - ▶ Gravity / Momentum
 - ▶ Falls (especially corners)
 - ▶ Buttons
- ▶ Interfaces
 - ▶ Cables
 - ▶ Displays
 - ▶ Buttons
- ▶ Minimize all else!

Flange

What is a **flange**?

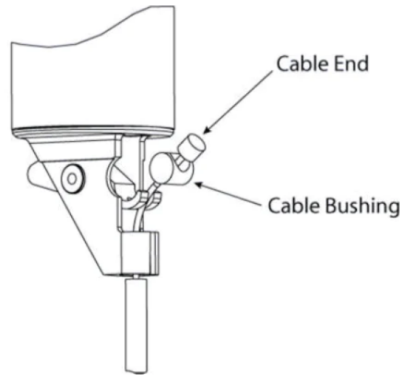
- ▶ Internal/external ridge/lip/rim
- ▶ ↑ **strength** / distribute **contact force** between mating pieces
- ▶ Commonly uses a **bolt circle** (BC)



BME

- ▶ Demo
- ▶ Tips:
 - ▶ Rename steps to build the part
 - ▶ Use variables for dimensions that may change
 - ▶ Use patterns when possible
 - ▶ Note version history / tag specific revisions
 - ▶ Look at different views, including “sections”
 - ▶ Create mechanical drawing

- ▶ What is a **bushing**?
 - ▶ It is actually a **bearing**.
 - ▶ Reduces friction between a rotating shaft and a fixed support member.
 - ▶ Typically made of metal or plastic, with lubrication.
- ▶ Other types of bearings:
 - ▶ Ball bearings
 - ▶ Roller bearings



Lab This Week

1. Create flange specified in the mechanical drawing
 - 1.1 Add a 5 mm hex head set screw (other specifics, e.g., thread pitch, are arbitrary) to the flange collar to hold a rod in place
 - 1.2 Generate an assembly with a mocked rod part
 - 1.3 Generate a mechanical drawing based on your part
2. Create the part and mechanical drawing for the Fox bushing

Please submit everything through Gradescope.